

Zakriya Asif Paracha

Software Engineer

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SUMMARY

Software Engineer with backend and data engineering experience, most recently at Motive building APIs and Apache Airflow ETL pipelines that turned raw panel data into compliance insights. Working toward full-stack, product-minded engineering. Computer Engineering background. Currently open to new opportunities.

EXPERIENCE

L2 Software Engineer at **Motive**

Sep 2024 – Aug 2026

Ruby on Rails · Python · Apache Airflow · SQL · PostgreSQL · Redis · Git · REST APIs

- Owned a complex Apache Airflow pipeline (Python, SQL) that ingested and transformed USA CSA panel data into clear, user-facing compliance insights.
- Set up ETL DAGs for a Canada-based data environment, extending pipeline coverage to a new region.
- Built and maintained backend services in Ruby handling production traffic with high reliability.
- Debugged and resolved critical production issues, reducing downtime.
- Designed and implemented RESTful APIs for internal services.
- Optimized PostgreSQL/Redis query performance and caching strategies.
- Contributed to code reviews and deployment practice improvements.

Embedded Systems Intern at **RISETech**

Jul 2023 – Sep 2023

Nordic nRF52 · Zephyr RTOS · BLE · SAADC · C/C++

- Worked on an IoT healthcare device using the Nordic nRF52 DK with Zephyr RTOS.
- Integrated SAADC to acquire EMG sensor data, transmitted over BLE.
- Developed firmware to manage the MPU6050 sensor over I2C.
- Optimized embedded algorithms for performance and memory efficiency on ARM microcontrollers.
- Supervised and mentored junior interns.

Software Intern at **BCUBE**

Sep 2021 – Oct 2021

- Developed Python software to control a biometric device with a custom UI.

Assembly Language Tutor at Fiverr

Mar 2021 – May 2023

MIPS · x86 · ARM Assembly · Debugging

- Delivered 120+ projects, maintaining a 5-star rating across 50+ clients.

EDUCATION

BE Computer Engineering, NUST, CEME, Islamabad

Graduated Jun 2024

CGPA 3.66 / 4.0

- 30+ Coursera courses, including specializations in Development of Secure Embedded Systems (EIT Digital), Introduction to Programming the IoT (UC Irvine), and Computer Vision with Embedded ML (Edge Impulse).

SKILLS

LANGUAGES

C/C++, Python, Ruby, SQL, Go

BACKEND & DATABASES

Ruby on Rails, Flask, REST APIs, PostgreSQL, Redis

DATA ENGINEERING

Apache Airflow, ETL Pipelines, DAG Design, SQL, Data Ingestion & Transformation

CORE CS

Operating Systems, System Programming, System Design, Data Structures & Algorithms

TOOLS

Git, Linux, Docker

ARCHITECTURE

RISC-V, Computer Architecture, CPU Design Fundamentals

EMBEDDED & SYSTEMS (HOBBY)

Embedded C, RTOS (Zephyr), Embedded Linux Basics, Firmware Development

HARDWARE PLATFORMS (HOBBY)

Arduino, ESP8266/32, Raspberry Pi, Nordic nRF52, Jetson Nano

LOW-LEVEL (HOBBY)

Interrupt Handling, Memory Management,
Boot Process Basics, I2C/SPI/UART

SELECTED PROJECTS

Optimizing Warehouse Item Transportation using Multi-Hop Robotic Networks (FYP)

2024

Python · Angular · MQTT · ESP8266

- Built a full-stack system for warehouse fleet coordination: a Python backend for multi-hop route optimization and collision avoidance, plus an Angular dashboard for real-time control and analytics.

Braille to Text Translator

2023

Python · Image Processing

- Built a Python image-processing tool that translates photographed Braille patterns into text.

C++ Audio Player

2021

C++ · WinForms

- Built a desktop audio player using WinForms and dirent.h.